Phys. 2110 HW 14 : 14. 17, 23, 24, 30, 43, 51, 61, 62, 79

14. 17 $P - P_{air} = \rho g h = (1.00 \times 10^3\ kg/m^3)(9.80\ m/s^2)(6.1\ m) = 6.0 \times 10^4\ Pa$

14. 23 (a) $P - P_0 = \frac{W}{A} = \frac{45.0\ N}{0.0707\ m^2} = 636\ Pa$

(b) The increase in pressure produces a force on the disk equal to the increase in weight. By Pascal's law the increase in pressure is transmitted to all points in the oil.

(i) $\Delta P = \frac{83.0\ N}{0.0707\ m^2} = 1170\ Pa$ (ii) $1170\ Pa$

14. 24 $\frac{F_2}{A_2} = \frac{F_1}{A_1}$, $A = \frac{\pi D^2}{4}$

$mg = F_2 = \frac{A_2}{A_1} F_1 = \frac{\pi D_2^2/4}{\pi D_1^2/4} F_1$

$\frac{D_2}{D_1} = \sqrt{\frac{mg}{F_1}} = \sqrt{\frac{(1520\ kg)(9.80\ m/s^2)}{125\ N}} = 10.9$

14.30 (a) $B = (1000\ kg/m^3)(0.650\ m^3)(9.80\ m/s^2) = 6.37 \times 10^3\ N$

(b) $B = T + mg$ $m = \frac{B - T}{g} = \frac{6.37 \times 10^3\ N - 900\ N}{9.80\ m/s^2} = 558\ kg$

(c) Now $B = \rho_{water} V_{sub}\, g$. V_{sub} is the volume of the sphere that is submerged.

$B = mg$ $\rho_{water} V_{sub} = m$

$V_{sub} = \frac{m}{\rho_{water}} = \frac{558\ kg}{1000\ kg/m^3} = 0.558\ m^3$

$\frac{V_{sub}}{V} = \frac{0.558\ m^3}{0.650\ m^3} = 85.8\%$

14.43 $P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$

$P_2 - P_1 = \frac{1}{2}\rho(v_1^2 - v_2^2) = \frac{1}{2}(1.20\ kg/m^3)((70.0\ m/s)^2 - (60.0\ m/s)^2) = 780\ Pa$

The net force exerted by the air is

$P_2 A - P_1 A = (780\ Pa)(16.2\ m^2) = 12600\ N$

The net force is upward.

14.51 (a) The length of the dam is L. $A = HL$

$$dF = p\,dA = \rho g h L\,dh$$

$$F = \int_0^H dF = \rho g L \int_0^H h\,dh = \rho g L \frac{H^2}{2} = \frac{\rho g A H}{2}$$

(b) The torque about the bottom on a strip of vertical thickness dh is

$$d\tau = dF(H-h) = \rho g L h (H-h)\,dh$$

$$\tau = \int_0^H d\tau = \rho g L \int_0^H h(H-h)\,dh = \rho g L \frac{H^3}{6} = \frac{\rho g A H^2}{6}$$

(c) The force depends on the width and on the square of the depth, and the torque depends on the width and the cube of the depth. The surface ~~of~~ area of the lake does not affect either result.

14.61 (a) The buoyancy force $B = \rho_w V_{sub} g$

$$\Sigma F_y = 0 \qquad B - mg = 0 \qquad \rho_w V_{sub} g - mg = 0$$

$$V_{sub} = \frac{m}{\rho_w} = \frac{900\,kg}{1000\,kg/m^3} = 0.900\,m^3$$

$$\frac{V_{sub}}{V_{obj}} = \frac{0.900}{3.0} = 30\%$$

(b) When the car is full submerged $V_{sub} = V$

$$B = \rho_w V g = (1000\,kg/m^3)(3.0\,m^3)(9.80\,m/s^2) = 2.94\times10^4\,N$$

The weight of the car $mg = (900\,kg)(9.80\,m/s^2) = 8820\,N$

Thus the weight of the water in the car when it sinks is

$$m_w g = B - mg \qquad m_w = 2.10\times10^3\,kg$$

$$V_w = \frac{m_w}{\rho_w} = \frac{2.10\times10^3\,kg}{1000\,kg/m^3} = 2.1\,m^3$$

$$\frac{2.1\,m^3}{3.0\,m^3} = 0.70 = 70\%$$

14.62 (a) $B = \rho_w V_{sub} g$ $\quad \Sigma F = 0 \quad$ $\rho_w V_{sub} g = m_{ice} g$

The volume displaced $V_{sub} = \frac{m_{ice}}{\rho_w} = \frac{9.70 g}{1.00 g/cm^3} = 9.70 cm^3$

(b) No; when melted, the cube produces the same volume of water as was displaced by the floating cube, and the water level does not change.

(c) $\frac{9.70 g}{1.05 g/cm^3} = 9.24 cm^3$

(d) The melted water takes up more volume than the salt water displaced, and so $0.46 cm^3$ flows over.

14.79 The volume of the anchor is $V_{anchor} = m/\rho = 4.453 \times 10^{-3} m^3$.
The barge rises in the water a vertical distance h, $hA = 4.453 \times 10^{-3} m^3$.
A is the area of the bottom of the barge.

$h = \frac{4.453 \times 10^{-3} m^3}{8.00 m^2} = 5.57 \times 10^{-4} m.$